Calcurion Maths Worksheets

GCSE Mathematics

Topic: Averages & Charts

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Find the mean of $4, 7, 9, 12, 8$. **[2]**

2. Find the median of $3, 8, 5, 2, 9, 6, 4$. **[2]**

3. Find the mode of $2, 5, 3, 5, 7, 5, 1$. **[1]**

4. Find the range of $12, 5, 18, 7, 9$. **[2]**

5. The mean of five numbers is 12. Four of the numbers are $10, 14, 8, 15$. Find the fifth number. **[3]**

6. Find the median of $4, 9, 2, 7, 5, 11$. **[2]**

7. A data set has mean 20 and 8 values. Find the total of the values. **[2]**

8. Find the mode and range of $3, 7, 3, 8, 5, 3, 9$. **[3]**

9. The heights (cm) of 6 students are $150, 162, 155, 170, 158, 165$. Find the mean height. **[3]**

10. In a frequency table, the scores $1, 2, 3, 4$ have frequencies $5, 8, 6, 1$. Find the mean. **[4]**

11. Find the median from the frequency table: scores 1–5 with frequencies $2, 4, 7, 5, 2$. **[3]**

12. A pie chart has 8 equal sectors. What angle does each sector represent? **[2]**

13. In a pie chart, a category has frequency 15 out of total 60. Find the angle for that category. **[3]**

14. The mean of x, $x + 2$ and $x + 4$ is 10. Find x. **[3]**

15. Find the interquartile range of the data: $2, 4, 5, 7, 8, 9, 11, 13, 15$. **[3]**

Section B – Hard (Q16–25)

16. The mean of 10 numbers is 15. One number is removed and the new mean is 14. What number was removed? **[4]**

17. A frequency table shows ages 10–14 with frequencies $3, 5, 8, 6, 4$. Estimate the mean age. **[5]**

18. The median of a set of 9 numbers is 12. What does this tell you about the data? **[2]**

19. In a histogram, the frequency density for a class of width 5 is 4. Find the frequency. **[2]**

20. The cumulative frequency for the upper quartile is 75% of the total. If there are 40 values, what is the upper quartile position? **[2]**

21. Data: $5, 7, 8, 10, 12, 15, 18, 20$. Find the lower and upper quartiles. **[4]**

22. A pie chart represents 180 students. The angle for "Maths" is $72°$. How many students chose Maths? **[3]**

23. Two sets of data have means 10 and 14 with 5 and 3 values respectively. Find the combined mean. **[4]**

24. A stem-and-leaf diagram has stems 1–3 and leaves showing 12 values. The median is the average of the 6th and 7th values. If these are 18 and 21, find the median. **[2]**

25. Explain the difference between the mean and the median, and when the median might be preferred. **[4]**

End of Worksheet • *Show all working clearly.*